

Long-Short Prompt Memory (LSPM)

The $\oplus$ Prostate



Prompt T_i

Fusion

Short Prompt Memory

T_i^{short}

EMA

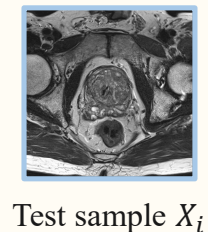
Long Prompt Memory

T_i^{long}

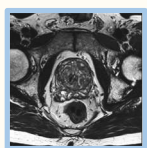
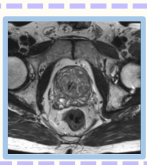
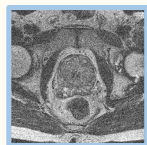
SAM3
Text
Encoder
 f_t

t_i^{short}

t_i^{long}



Test sample X_i



K augmentations

SAM3
Image
Encoder
 f_v

$\{z_i^0, \dots, z_i^k, \dots, z_i^K\}$

z_i^*

SAM3
Detector
 f_d



Trainable Parameters



Frozen Parameters

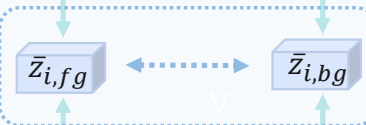
Original Inference

Pseudo Label Inference

Concept Alignment Contrast (CAC)

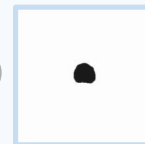
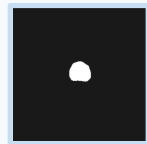
t_i^{short}

Cosine Similarity



C_i^k

CAC Score for
view X_i^k



$\mathcal{L}_c$

$\mathcal{L}_e$

$\mathcal{L}_d$



$\mathcal{L}_d$ Dense Supervision Loss

$\mathcal{L}_c$ Contrast Loss

$\mathcal{L}_e$ Entropy Loss

Densely Supervised Prompt Update (DSPU)